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Research Article

Explicable Machine Learning for Predicting High-Efficiency Lignocellulose Pretreatment Solvents Based on Kamlet−Taft and Polarity Parameters

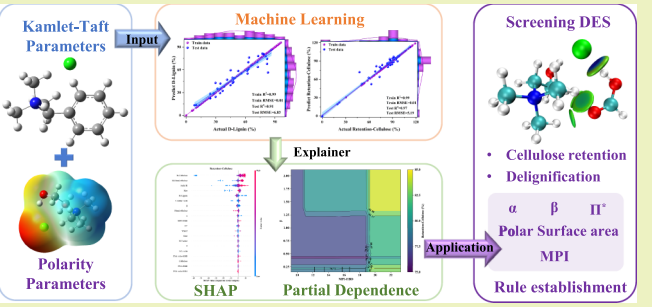
Hanwen Ge, Yuekun Bai, Rui Zhou, Yaoze Liu, Jiahui Wei, Shenglin Wang, Bin Li, and Huanfei Xu*

Cite This: ACS Sustainable Chem. Eng. 2024, 12, 7578−7590

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ABSTRACT: Incorporating density functional theory (DFT) and machine learning (ML) methodologies, an intrinsic relationship model was developed utilizing the Kamlet−Taft parameters and polarity values of 104 deep eutectic solvents (DES). DES with high lignocellulosic pretreatment efficiency were expected to be screened through the synergistic combination of hydrogen bond acidity (α), hydrogen bond basicity (β), polarization (Π^*) and molecular polarity index (MPI). Partial least-squares (PLS) models and a variety of ML models were used to predict cellulose retention and delignification. The XGBoost model has the highest predictive performance with R^2 of 0.97 and 0.91, respectively. Feature importance analysis and partial dependence analysis were used to explain the importance of variables based on the XGBoost model. Feature importance analysis showed that α, β, Π^* of DES and MPI of hydrogen bond donor determined the pretreatment efficiency. The partial dependence analysis showed that the relationship among 4 parameters and the pretreatment efficiency is nonlinear, and there are multiple extreme values in different intervals. The model gave a parameter range corresponding to the high pretreatment efficiency. Based on the range of 4 parameters given in this study, the composition and ratio of DES can be selected to ensure that at least 80% of the cellulose is retained and 50% of the lignin is removed. Molecular simulation results showed that these highly efficient DES often contain a large number of hydrogen bonds and highly polar groups.



KEYWORDS: *lignocellulose, deep eutectic solvents, pretreatment, machine learning, solvent properties*

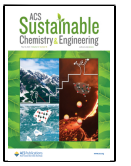
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1. INTRODUCTION

Lignocellulosic biomass can be obtained from nature, and it is recognized as having the advantages of wide distribution, huge reserves, and renewable, so it has great value of high value-added conversion and utilization.[1] As a natural carbon-containing resource, it also has a major advantage in that the lignin component contains a benzene ring structure, which determines its potential for high-value utilization.[2] The cellulose, hemicellulose, and lignin in lignocellulosic biomass can be separated by pretreatment. The separated components can be used in energy, electrode materials, membrane materials, flexible materials, and other high-value-added materials.[3−5] These biomass components also can be further synthesized into platform compounds by chemical modification or enzymatic hydrolysis.[6,7] Among many pretreatment methods, DES has high pretreatment efficiency in the pretreatment process of lignocellulosic biomass, which has attracted much attention from researchers.[8]

DES is composed of hydrogen bond donor (HBD) and hydrogen bond acceptor (HBA), which has unique solvent properties due to its strong hydrogen bond system.[9] Among the many solvent properties, the Kamlet−Taft (K-T) parameters and polarity parameters of DES have attracted much attention.[10,11] The K-T parameter contains α, β, and Π^*. Several studies have explored the relationship between the values of the K-T parameters and the separation efficiency of cellulose, hemicellulose, and lignin. Based on the conclusion of the experiment, the structure−activity relationship is established, and the conclusion of the relationship is usually to conclude that α, β, or $\beta-\alpha$ is positively correlated with pretreatment efficiency.[12] There are limitations in employing a straightforward linear relationship to evaluate the K-T parameters and the pretreatment efficiency of DES. A prior study aimed to establish a machine learning model for analyzing the relationship between K-T parameters and lignin dissolution, validating the above viewpoint.[13] Therefore, these relationships need to be further analyzed and sorted out. Hydrogen bonds are noteworthy molecular interactions of

Received: February 25, 2024
Revised: April 12, 2024
Accepted: April 15, 2024
Published: April 29, 2024



ACS Publications

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https://doi.org/10.1021/acssuschemeng.4c01563
ACS Sustainable Chem. Eng. 2024, 12, 7578−7590

DES, and the composition of these hydrogen bonds is thought to be influenced by the polarity parameters of the molecule itself.[14] Therefore, it is more accurate to use the combination of the solvent polarity parameter and K-T parameter to construct the correlation model between the solvent and DES pretreatment efficiency. Among the polarity parameters, the MPI and the polar surface area of the solvent were particularly concerned, which are considered to directly affect the efficiency of DES pretreatment.[15,16] MPI serves as a significant parameter for assessing the polarity of molecules through the analysis of the electrostatic potential distribution on the molecular surface. The polar surface area of a molecule provides a concise gauge of its hydrogen bonding capacity.[17] In fact, there is a relationship between the K-T parameter and the polarity parameter. The relationship between solvent parameters and pretreatment efficiency was analyzed by establishing a model based on the above solvent parameters. Therefore, it is of great research value to determine a mechanism for screening efficient DES.

Machine learning, as a new tool in the field of artificial intelligence, can analyze complex and multiple variables, and then get the internal relationship.[18] In recent years, ML has been widely employed in the field of high-value utilization of lignocellulosic biomass.[19,20] The molecular simulation results can be input into the ML model as descriptors to better analyze the interaction mechanism between solvent and biomass.[21,22] This provides a method to combine the macroscopic variables measured in experiment with the microscopic variables calculated in quantum chemistry to analyze the novel internal relationship among those variables.[23] Several studies are dedicated to developing databases and models for predicting solvent properties, offering crucial theoretical and data support for the design and development of novel DES.[24,25] The results of ML model output were interpreted through feature importance analysis and partial dependence analysis, and the importance of variables is ranked.[26] Some studies also analyze the internal relationship of variables in the DES pretreatment biomass process through multivariate analysis methods, such as PLS model.[27] These conclusions can guide the synthesis of highly efficient pretreated DES and have important significance for the industrial application of DES in the future.

In this study, based on the pretreatment process of lignocellulosic biomass by DES, the macrovariables measured by experiments such as lignocellulosic and solvent components, pretreatment conditions, and K-T parameters, as well as microvariables such as polarity parameters calculated by quantum chemistry were used as the data set. The PLS model was used to analyze the relationship between the variables in the data set and the cellulose retention rate and delignification. The data set was input into multiple ML models, and the output of ML models was interpreted by means of feature importance and partial dependence analysis. The variable relationships between K-T parameters, polarity parameters, and pretreatment efficiency were explained, and the importance of variables in the process was analyzed. Based on the K-T parameters of DES and MPI of HBD, a DES screening model for efficient pretreatment of lignocellulose was constructed, and the range of parameters was determined. Finally, the interaction mechanism of DES was verified by a molecular simulation. This study provides novel insights for the study of the DES efficient pretreatment of lignocellulose.

2. MODELS AND METHODS

2.1. Data Collection and Descriptor Calculation. To ensure the diversity of DES included in the data set, 20 articles were collected from ScienceDirect and Web of Science literature databases. The collected literature must focus on the pretreatment of lignocellulosic biomass by DES, and the K-T parameters of the DES used should be calculated. The data set contains a total of 18 HBA and 23 HBD, which together involve 104 DES composed of them. Eighteen different lignocellulosic biomass feedstocks were pretreated with the above DES. Details of these different types of DES and lignocellulosic biomass can be consulted from the Supporting Information.

The variables in the data set were divided into seven groups, composition of biomass, composition of DES, solvent polarity parameters, solvent K-T parameters, pretreatment conditions, composition of pretreated biomass, and pretreatment efficiency. The details of the variables were placed in Table 1. To ensure data consistency, the solvents used for determination of all K-T parameters were specified as Nile red, 4-nitroaniline and N, N-diethyl-4-nitroaniline. For example, MPI, molecular polar surface area, and proportion of different HBA and HBD contained in the solvent polarity parameters variable group. Among these, the calculation formula for the MPI is depicted in formula 1.[28] Gaussian software was used to perform the molecular structure optimization tasks for HBA and HBD at the B3LYP-D3 (BJ)/def2-SVP level, and then the energy optimization tasks were performed at the B3LYP-D3 (BJ)/def2-TZVP level.[29] The obtained files recording its structure and energy information were input into Multiwfn for the calculation of polarity parameters.[30] R Factor, a variable to measure the severity of reaction, was added to the group of preconditioned variables, and its calculation method was shown in formula 2.[31]

$$\mathrm{MPI} = (1/A)\iint_S |V(r)|\mathrm{d}S \tag{1}$$

where A represents the molecular surface area; V represents the molecular electrostatic potential; and S represents the molecular surface.

$$\text{R Factor} = \log(t \times \exp(T - 100)/14.7) \tag{2}$$

where t represents the pretreatment time; T represents the pretreatment temperature.

2.2. Partial Least-Squares Method. The PLS model was used to explore the relationship between the two matrices of the independent variable and dependent variable to obtain the maximum explanation of the spatial variance of the dependent variable. In this study, Retention Cellulose and D-Lignin in the data set were set as dependent variables, and the remaining 26 variables were set as independent variables. The PLS model in data analysis software SIMCA-P 16 is used for multivariate analysis of data sets. Before that, the data set needed to be entered into the software in the form of an Excel spreadsheet. Some of the data need to be averaged to ensure the normal operation of software functions. Components in the model were adjusted to ensure maximum explanatory and predictive performance of the model.

2.3. ML Model Prediction and Evaluation. In this study, five ML models were constructed and evaluated, specifically Extreme Gradient Boosting (XGBoost), Random Forest (RF), Gaussian Process Regression (GPR), Support Vector Regression (SVR), and K-Neighbor Regression (KNR). These models proved to have high predictive power even when trained on relatively small databases.[32] The above ML models were used to predict dependent variables one by one. The setting of variables in the model is shown in Table 1. The source codes for the model were obtained by loading the model in the sklearn package and executed in the Python 3.10.11 operating environment. The hold-out method is applied to divide the data in the database. 80% of the data is used for the training of the model, and 20% is used for the testing of the model. The Grid Search CV function is used to adjust the hyperparameters of the model to get the model with the best predictive performance. The hyperparameter combinations of all optimization models are listed in Table 2.

https://doi.org/10.1021/acssuschemeng.4c01563
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Table 1. Detailed List of the Variables Contained in the Models[a]

groups	abbreviation of variables	full name of variables	PLS model	ML model
composition of biomass	size	particle size of raw biomass, mesh	X	X
	cellulose	cellulose content of raw biomass, %	X	X
	hemicellulose	hemicellulose content of raw biomass, %	X	X
	lignin	lignin content of raw biomass, %	X	X
composition of DES	1-molar ratio	the molar amount of the first HBD in DES	X	X
	2-molar ratio	the molar amount of the second HBD in DES	X	X
	water	mass ratio of water added to DES	X	X
	metal	whether metal components are added to DES	X	X
	third-C	whether DES is a ternary DES	X	X
solvent polarity parameters	MPI-HBA	MPI of HBA, kcal/mol	X	X
	PSA-HBA	polar surface area of HBA, Å^2	X	X
	PSA ratio-HBA	polar surface area ratio of HBA, %	X	X
	MPI-HBD	MPI of HBD, kcal/mol	X	X
	PSA-HBD	polar surface area of HBD, Å^2	X	X
	PSA ratio-HBD	polar surface area ratio of HBD, %	X	X
solvent K-T parameters	α	hydrogen bond acidity of DES	X	X
	β	hydrogen bond basicity of DES	X	X
	Π^*	dipole/polarization of DES	X	X
pretreatment conditions	T	pretreatment temperature, °C	X	X
	time	pretreatment time, h	X	X
	R factor	severity factors of pretreatment reaction	X	X
	S/L ratio	mass ratio of DES to raw biomass	X	X
composition of pretreated biomass	Solid R	solid recovery rate after pretreatment, %	X	X
	R-cellulose	cellulose content after pretreatment, %	X	X
	R-hemicellulose	hemicellulose content after pretreatment, %	X	X
	R-lignin	lignin content after pretreatment, %	X	X
pretreatment efficiency	retention-cellulose	cellulose retention rate after pretreatment, %	Y	Y
	D-Lignin	delignification rate after pretreatment, %	Y	Y

[a]X, independent variables; Y, dependent variables.

Table 2. Detail List of the Hyperparameters of the Optimization model

model	variable	hyperparameter	value
XGBoost	retention-cellulose	max_depth	3
		learning_rate	0.73
		n_estimators	300
	D-lignin	max_depth	5
		learning_rate	0.35
		n_estimators	300
RF	retention-cellulose	n_estimators	46
		max_depth	9
	D-lignin	n_estimators	47
		max_depth	10
GPR	retention-cellulose	alpha	5
		n_restarts_optimizer	20
	D-lignin	alpha	20
		n_restarts_optimizer	20
SVR	retention-cellulose	C	100
		gamma	0.0001
		epsilon	0.01
	D-lignin	C	100
		gamma	0.0003
		epsilon	0.01
KNR	retention-cellulose	n_neighbors	1
	D-lignin	n_neighbors	1

All models were configured with appropriate evaluation parameters to determine their reliability, which also served as an indicator of whether the hyperparameters needed adjustment. The performance of the train set and the test set was evaluated using several metrics, including the determination coefficient (R^2), mean square error (MSE), root-mean-square error (RMSE), mean absolute error (MAE), and absolute average relative deviation percent (AARD%). Its calculation formula is shown in formulas 3−7:

$$R^2 = \left(\sum_{i=1}^{n}\left(y_{\text{actual}} - \overline{y}\right)^2 - \sum_{i=1}^{n}\left(y_{\text{actual}} - y_{\text{predict}}\right)^2\right) \Big/ \sum_{i=1}^{n}\left(y_{\text{predict}} - \overline{y}\right)^2 \quad (3)$$

$$\text{MSE} = \sum_{i=1}^{n}\left(y_{\text{actual}} - y_{\text{predict}}\right)^2 / n \quad (4)$$

$$\text{RMSE} = \sqrt{\sum_{i=1}^{n}\left(y_{\text{actual}} - y_{\text{predict}}\right)^2 / n} \quad (5)$$

$$\text{MAE} = \sum_{i=1}^{n}\left|y_{\text{actual}} - y_{\text{predict}}\right| / n \quad (6)$$

$$\text{AARD}(\%) = \sum_{i=1}^{n}\left|\left(y_{\text{actual}} - y_{\text{predict}}\right)/y_{\text{actual}}\right| / n \times 100 \quad (7)$$

where n represents the number of total sample points; y_{actual} represents the actual values; y_{predict} represents the predicted values; $\overline{y}$ represents the sample average values.

Based on the ML model with optimal performance, the accuracy of the prediction performance of the target variable was further analyzed. The relative deviation of each target variable was calculated. In addition, the application domain of the model was further determined, and the reliability of the prediction values of the model within the application domain was determined.[21] The formula for calculating the leverage value and warning value in the application domain is shown in formulas 8and9:[33]

$$h_i = x_i^{\mathrm{T}}\left(X^{\mathrm{T}}X\right)^{-1}x_i \quad (8)$$

$$h^* = 3(p+1)/n \quad (9)$$

where h_i represents the lever value; h^* represents the warning value; x_i represents the row vector about descriptor; X represents the matrix

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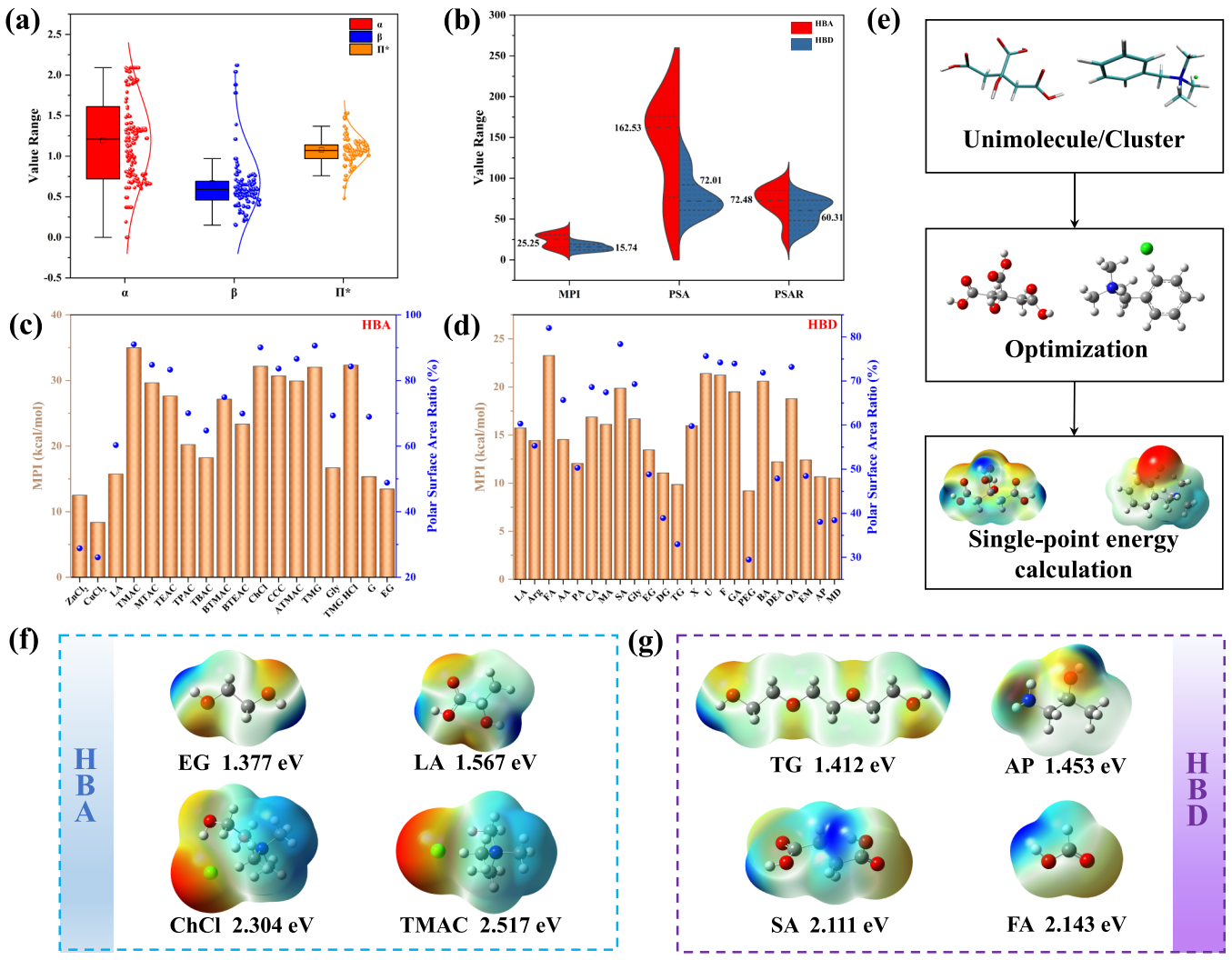


Figure 1. (a) Statistical analysis of the input Kamlet−Taft parameters; (b) violin statistical plot of the input polarity parameters was obtained by DFT calculating; (c) polarity parameter of the HBA in the data set; (d) polarity parameter of the HBD in the data set; (e) molecular structure and energy optimization using DFT basis set and steps; (f) electrostatic potential plot of representative HBA molecules; (g) electrostatic potential plot of representative HBD molecules.

about descriptors; p represents the number of all descriptors; and n represents the number of data points.

2.4. Feature Importance Analysis. In order to increase the interpretability of the ML model, the feature importance analysis of all variables was carried out based on the model with optimal predictive performance. Based on the theory of SHapley Additive exPlanation (SHAP) and Feature Importance Score (F-Score), the dependent variable is taken as the target variable, and the feature importance analysis is carried out for each target variable individually. Each independent variable in the model will be assigned a SHAP value and F-Score, and then, they will be sorted based on their score to determine the variable that has the most impact on the model output. In addition, adding the scores of a class of variables by grouping can assess the effect of the group of variables on the dependent variable. The SHAP package is of version 0.43.0 and is executed in the Python 3.10.11 operating environment.

2.5. Partial Dependence Analysis. In order to comprehensively elucidate the impact of one or more independent variables on dependent variables. In this study, one-dimensional or two-dimensional partial dependence analysis were carried out for some variables with significant feature importance based on the model with optimal predictive performance. Through the partial dependence analysis, the influence of the value change of the independent variable in different ranges on the value of a dependent variable can be obtained so as to determine the reasonable selection range. DES screened through the above selection range, which had a better pretreatment ability, were obtained. The source code for the partial dependency analysis used in this study is from the PartialDependenceDisplay in the sklearn package.

2.6. Details of Quantum Chemical Calculations. In this study, the method of quantum chemistry calculation was used to verify the selected DES, and the microscopic mechanism of the DES was visualized. The DES cluster structure was constructed by molclus software.[34] The constructed cluster structure was optimized by the semiempirical method using MOPAC software at the PM6-DH+ level.[35] The objective of semiempirical optimization is to identify cluster structures with comparatively lower energy in order to enhance calculation accuracy and conserve computational resources. The structural optimization task for the cluster structure was carried out using the Gaussian software at the M062X/6-31G(d) level.[29,36] To conduct the optimization of energy tasks, the optimized structure was utilized, and the calculations were performed at the M062X/6-311+g(d,p) levels. The interaction energy between HBA and HBD was calculated after correcting for the Basis Set Superposition Error (BSSE), utilizing the keyword "counterpoise = 2".[37] In order to visualize the weak interaction between DES molecules, Multiwfn was used to analyze the optimized molecules using an independent gradient model based on the Hirshfeld partition (IGMH).[30,38] The

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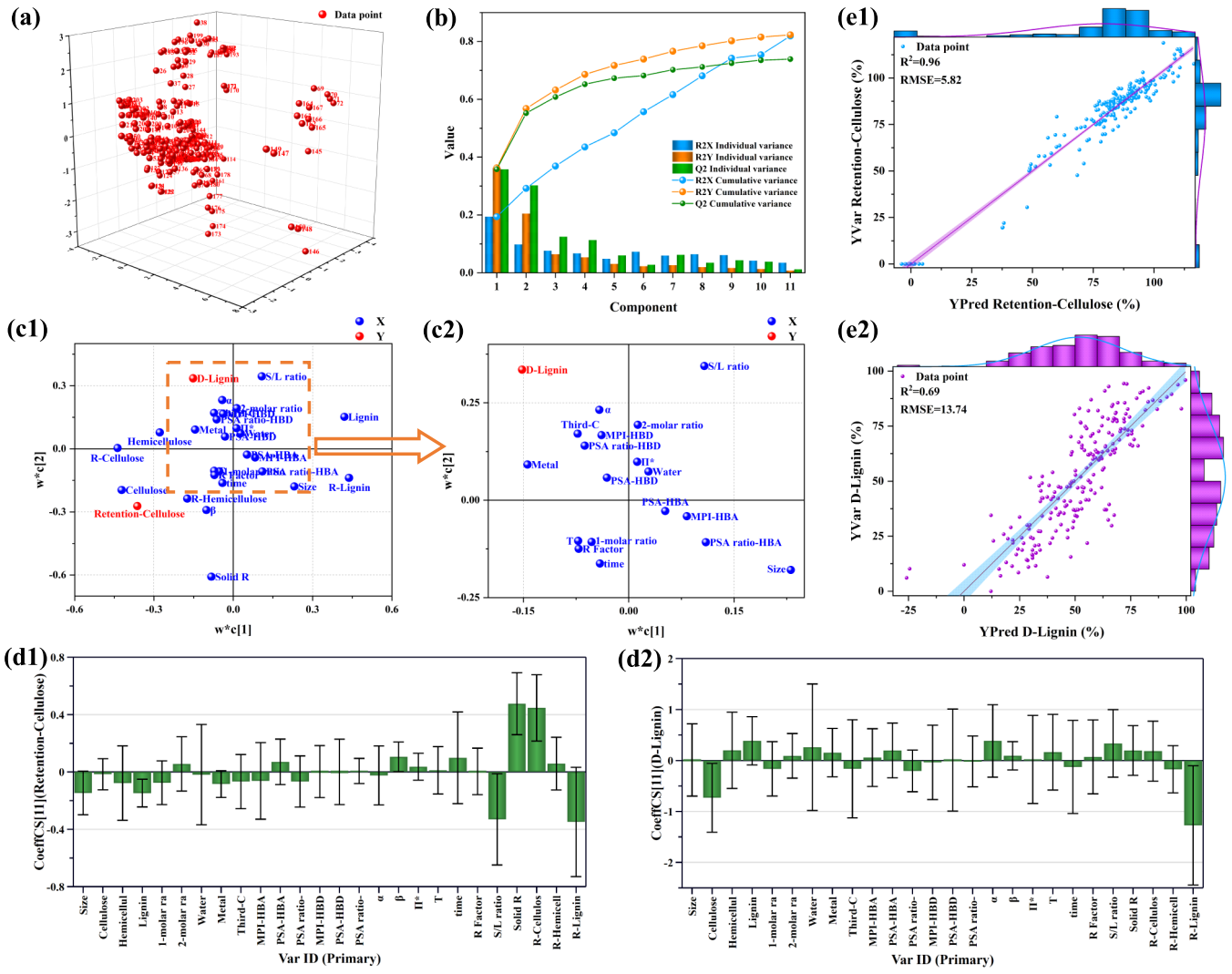


Figure 2. Multivariate analysis of the data set by PLS model: (a) 3D scatterplot of data set; (b) variances and cumulative variances of component; (c1, c2) loading scatterplot of variables; (d1, d2) coefficient plot of pretreatment efficiency variables; (e1, e2) marginal histogram of predictive ability of pretreatment efficiency variables.

HUMO-LUMO gap of DES molecules was calculated to roughly estimate the polarizability of DES molecules, and the calculation formula is as follows:

$$\text{HOMO} - \text{LUMOgap (eV)} = E_{\text{LUMO}} - E_{\text{HUMO}} \quad (10)$$

where E_{LUMO} represents the energy of the lowest unoccupied molecular orbital E_{HUMO} represents the energy of the highest occupied molecular orbital.

3. RESULTS AND DISCUSSION

3.1. Data Statistics and Variable Analysis. Analyzing the data set statistics enables a deeper comprehension of variable distributions and aids in understanding the characteristics of variables within the database.[39] The distribution of K-T parameters and polarity parameters of all DES in the database were statistically analyzed, and the results are shown in Figure 1a,b. In the K-T parameter of DES, the value of α ranges from 0.19 to 2.09 with an average of 1.19. The value of β ranges from 0.15 to 2.12 with a mean of 0.67. Moreover, the value of Π* ranges from 0.48 to 1.53 with an average of 1.08. The wide range of K-T parameters indicates the diversity of solvents included in the database. The effect of different hydrogen bond-forming abilities on pretreatment efficiency was further measured. The polarity parameters were calculated by Multiwfn software. Molecular polar surface area ratio refers to the ratio of the polar surface area of a molecule to the total surface area. The setting of this index is to avoid the influence of the size of the molecule volume on the value of the polar surface area. The calculation results of the polarity parameters of each HBA and HBD are shown in Figure 1c,d. The MPI of HBA ranges from 8.39 to 34.99, while the MPI of HBD ranges from 9.21 to 23.27. This means that most of the molecules included in the database have unevenly distributed charges and high polarity.[40] Among all HBA, tetramethylammonium chloride (TMAC), and choline chloride (ChCl) have larger MPI and molecular polar surface areas, while ethylene glycol (EG) and lactic acid (LA) have smaller MPI and molecular polar surface areas. Among all HBD, the larger MPI and polar surface areas were formic acid (FA) and Succinic Acid (SA), while the smaller MPI and polar surface areas were triethylene glycol (TG) and amino-2-propanol (AP). Some experimental results have reported that highly polar DES can improve the solubility of lignin and increase the fractionation efficiency of lignocellulosic biomass.[41,42]

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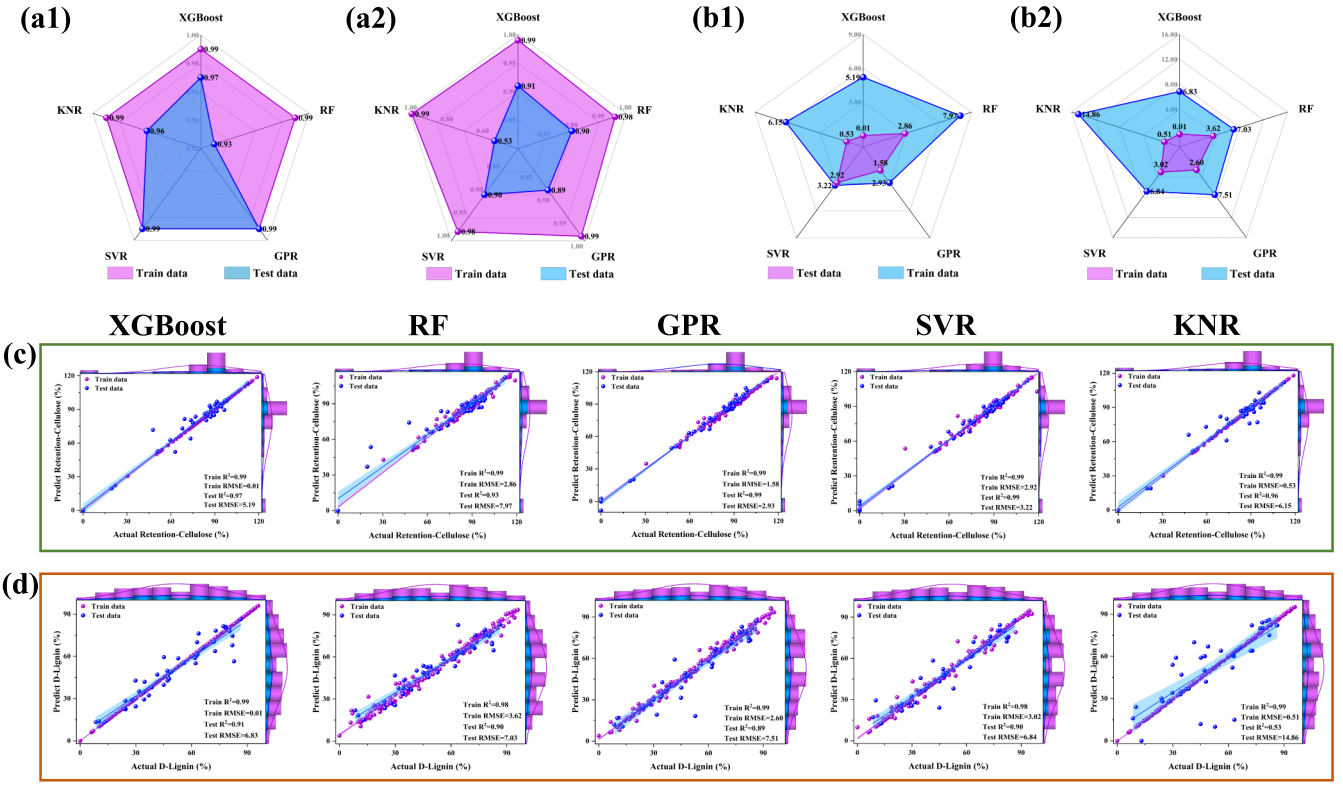


Figure 3. (a1, a2) Rader map of the R^2 performance for ML models; (b1, b2) Rader map of the RMSE performance for ML models; (c) comparison of retention-cellulose prediction results; (d) Comparison of D-Lignin prediction results.

The structure and energy of all molecules were optimized by Gaussian analysis before the polarity parameters were calculated. The optimization process and the basic set employed are displayed in Figure 1e. The electrostatic potential of the optimized molecule with greater polarity was calculated, and the calculation results are shown in Figure 1f. The red part of the diagram represents the negative electrostatic potential, and the blue part represents the positive electrostatic potential. The distribution of locations where weak interactions occur between functional groups can be comprehended by analyzing the electrostatic potential of the molecule.[43] The greater extreme value of the electrostatic potential of the molecule is represented by the greater polarity of the molecule. The negative extreme point of electrostatic potential in the highly polar HBA molecule appears near Cl^-. The positive extreme point of the electrostatic potential in HBD molecules with higher polarity appears near the hydroxyl group on the carboxyl group. In the formation of DES, the two are more likely to produce hydrogen bonds with stronger forces, which is explained by the above conclusion. This has also been reported in some previous studies of microscopic molecular mechanisms.[44] Interestingly, although many HBD molecules with larger molecular volumes have more hydroxyl groups, the polarity of the molecules is smaller. The longer carbon chain results in a smaller proportion of the polar surface area of the molecule, which is the reason for the above phenomenon. Similar conclusions have been reported in some studies of solvent properties in combination with molecular simulations and experiments.[45]

3.2. Key Variable Analysis Based on the PLS Model.

The relationship between the dependent variable and the independent variable can be well sorted out by the PLS model, and then the influence of the independent variable on the dependent variable can be measured.[46] In this study, the PLS model was used to conduct a multivariate analysis of DES pretreatment lignocellulosic biomass data set to find the key variables affecting the cellulose retention rate and delignification rate. The PLS model has the best explanatory and predictive performance in 11 components. The cumulative explanatory variance R2X of the independent variable is 0.819, the cumulative explanatory variance of the dependent variable is 0.823, and the cumulative explanatory variance Q2 is 0.739. This means that the model can explain more than 82% of the variables and predict about 74% of the information. The output details of the model are shown in Figure 2. The data were input to the model and have no strong outliers.

The scatterplot of variable relationships in the model is shown in Figure 2c. A stronger positive correlation is indicated by variables closer to each other within the same quadrant. Conversely, a stronger negative correlation is indicated by variables that are more distant from each other within the diagonal quadrant. In this model, it can be found that delignification is positively correlated with α and the polarity parameters of HBD, and negatively correlated with the polarity parameters of HBA. The data in the data set showed that ternary DES had better lignin removal effect than binary DES. By comparing the experimental results, it can be found that these ternary DES often add a certain amount of alcohol as HBD in the acidic DES system.[47,48] Delignification does not always increase with the increase of α, but existential extremum in some specific ranges, which is proved by the above phenomenon. The relationship between α and delignification is nonlinear, requiring determination of the specific range that corresponds to the maximum. There was a positive correlation between the cellulose retention rate and β, indicating that the increase of the β value of DES could enable more cellulose to

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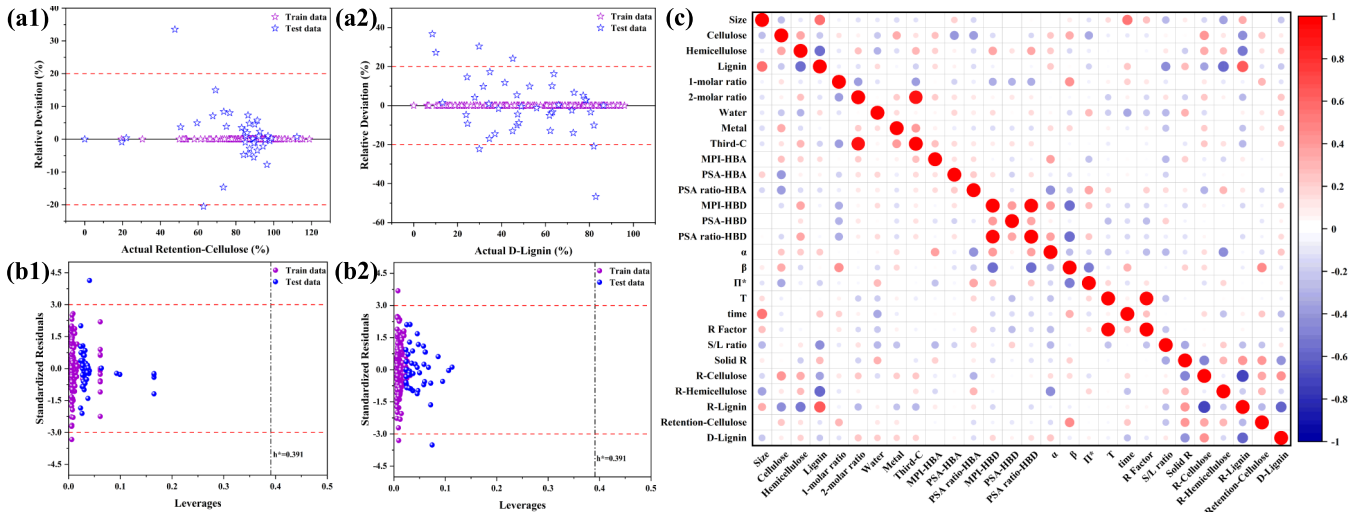


Figure 4. (a1) Relative deviation plot of Retention-Cellulose based on the XGBoost model; (a2) relative deviation plot of D-Lignin based on the XGBoost model; (b1) Williams plot of Retention-Cellulose based on the XGBoost model; (b2) Williams plot of D-Lignin based on the XGBoost model; (c) coefficient plot of the spearman correlation matrix.

be retained. The above relevant conclusions can be summarized in the experimental phenomena reported in some studies.[47]

The correlation coefficients of dependent variables are displayed in Figure 2d. The variable of retention-cellulose exhibited positive correlations with β and pretreatment time, while negative correlations were observed with the solid−liquid ratio and the molar amount of HBD added. For the D-Lignin variable, α and solid−liquid ratio were positively correlated, while the negatively correlated variables were the proportion of the polar surface area of the HBA molecule and the molar amount of HBD added. Some studies have indicated that DES with higher α values has a better ability to provide hydrogen bonds, release more H^+, and ultimately lead to higher delignification.[12] The model results demonstrate a strong correlation between the value of α and the polarity parameter in DES. The PLS model was used to predict the two dependent variables. It can be seen from this that the R^2 and RMSE of the PLS model for the predicted value of cellulose retention are 0.96 and 5.82, which means the error is small. However, the model prediction performance of delignification variables is poor.

3.3. Models Predict Performance Analysis and Applicability Domain. With the development of ML theory and technology, its research in the field of high-value utilization of biomass has gradually increased. Some novel models have been developed to accommodate a more complex data set. In this study, five ML models including XGBoost, RF, GPR, SVR, and KNR were selected to predict the data set of the DES pretreatment process of biomass. The prediction results of different dependent variables are shown in Figure 3c,d. The prediction results are compared to select the model with a better prediction performance to ensure the accuracy of the prediction results. The comparison of model prediction performance is shown in Figure 3a,b in the form of radar charts. The prediction performance of the KNR model for the dependent variable of delignification is only 0.51, and the prediction performance of all models for the dependent variable is above 0.89. Among them, the prediction performance of the GPR model and SVR model for the dependent variable of cellulose retention rate can reach 0.99. The XGBoost model is deemed to be the most accurate model for achieving comprehensively precise prediction of two dependent variables. It demonstrates excellent application potential in predicting biomass pretreatment efficiency and solvent properties.[16,33] In the case of the cellulose retention rate, the model achieved an R^2 of 0.97 and an RMSE of 5.19. For delignification prediction, the R^2 was 0.91, and RMSE was 6.83. The above values presented by the RMSE are considered to be due to the wide data distribution range of the data in the data set and the large overall data values.[26,49]

The accuracy and reliability of the prediction results of the XGBoost model with optimal prediction performance were further verified to enhance the generalization ability of the conclusions of this study. The calculation of relative deviation of prediction results of target variables and the determination of application domain are considered effective tools to reduce the uncertainty of prediction data.[50] Compared with the train set, the AARD% of the test set increases to some extent. The AARD% values of the two target variables increased to 8.79 and 6.45%, respectively. Although a smaller value of AARD% indicates a higher accuracy of the model prediction, in several studies of precise models used to predict solvent properties, AARD% of the model has been reported to be similar to the above value.[51] The calculated results of relative deviations for different target variables are shown in Figure 4a1,a2. The results in the figure show that the relative variation of the training set is small and the training effect of the model is well verified. The relative deviation of the test set fluctuates to a certain extent, but the absolute value of the relative deviation is basically within 20%. Such prediction results are considered relatively accurate.[52] The leverage values and standardized residuals of the model predictions were calculated, and the Williams plot was drawn based on the above results to determine the application domain of the model. The above results are shown in Figure 4b1,b2. Data points where the absolute value of the standardized residual is greater than 3 or the leverage value is greater than the warning value are considered data outliers.[53] The number of data outliers of this model is very few, and most of the data points are in the application domain. Therefore, the data of the model are

https://doi.org/10.1021/acssuschemeng.4c01563
ACS Sustainable Chem. Eng. 2024, 12, 7578−7590

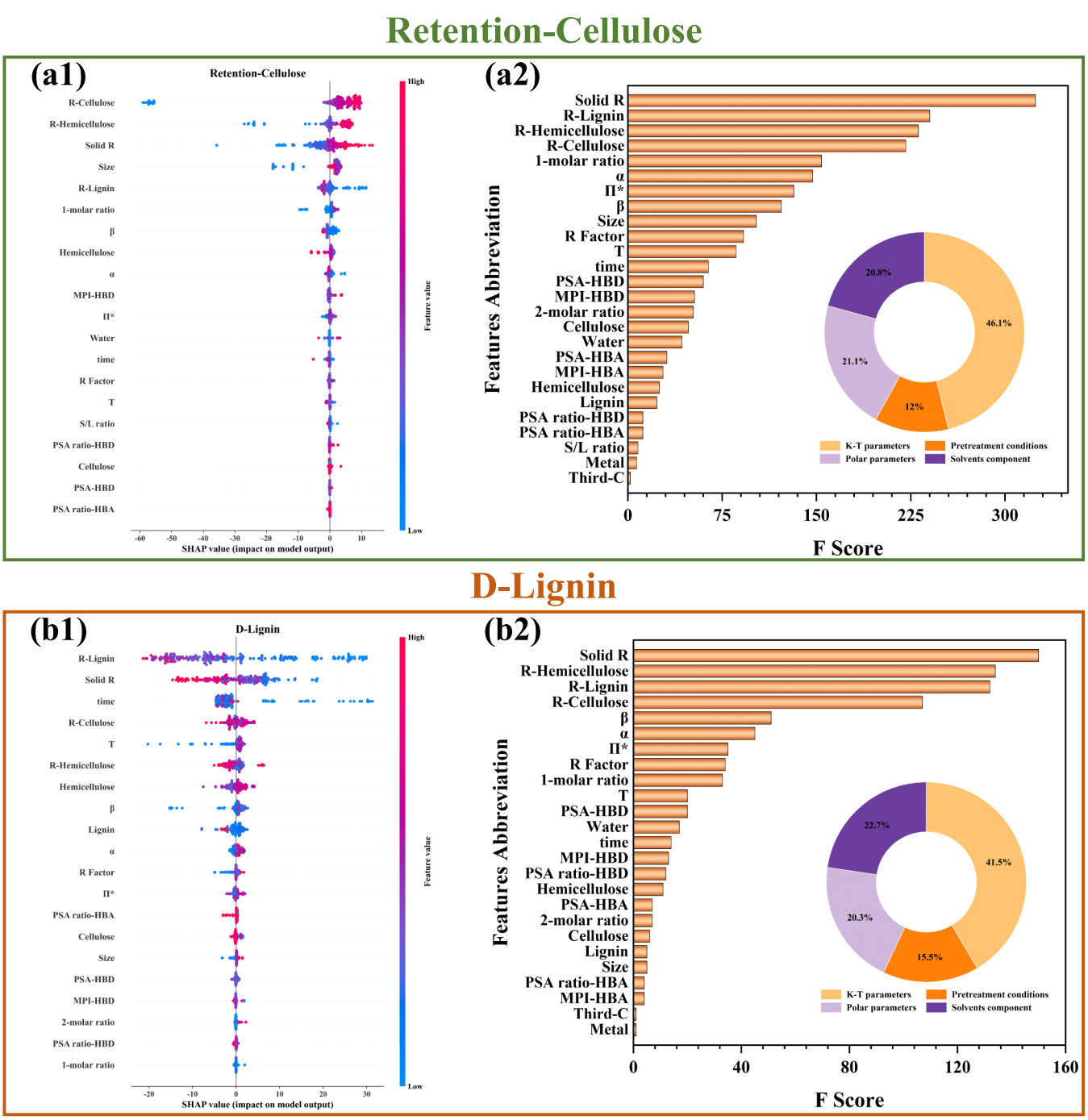


Figure 5. Feature importance analysis of pretreatment efficiency variables: (a1) variables SHAP value for retention-cellulose; (a2) variables F-Score for Retention-Cellulose; (b1) variables SHAP value for D-Lignin; and (b2) variables F-Score for D-Lignin.

reliable, and the conclusion can summarize the various types of DES involved in the model.

In order to sort out the correlation of variables in a complex nonlinear data set, the Spearman correlation matrix was used to analyze the data set. The correlation of the variables is shown in Figure 4c. The red dots in the matrix indicate positively correlated, and the blue dots indicate negatively correlated. It can be seen from the figure that the retention rate of cellulose is positively correlated with β and the delignification is also positively correlated with α. The polarity parameters of the solvent do not exhibit a direct correlation with the aforementioned dependent variables. However, α values were positively correlated with the MPI of HBA and HBD, while β values were negatively correlated with the MPI of HBD. Π* showed a negative correlation with the MPI of HBA and a positive correlation with the MPI of HBD. It can be revealed that there is an obvious correlation between the K-T parameters and the polarity parameters of the solvent. Interestingly, the α values were negatively correlated with the amount of hemicellulose after pretreatment. Several experimental studies have consistently reported comparable findings, although these outcomes appear to be influenced by the composition and properties of biomass feedstocks and solvents, resulting in some degree of uncertainty.[54]

3.4. Feature Importance Analysis Based on SHAP and F-Score. The application of ML to the pretreatment of

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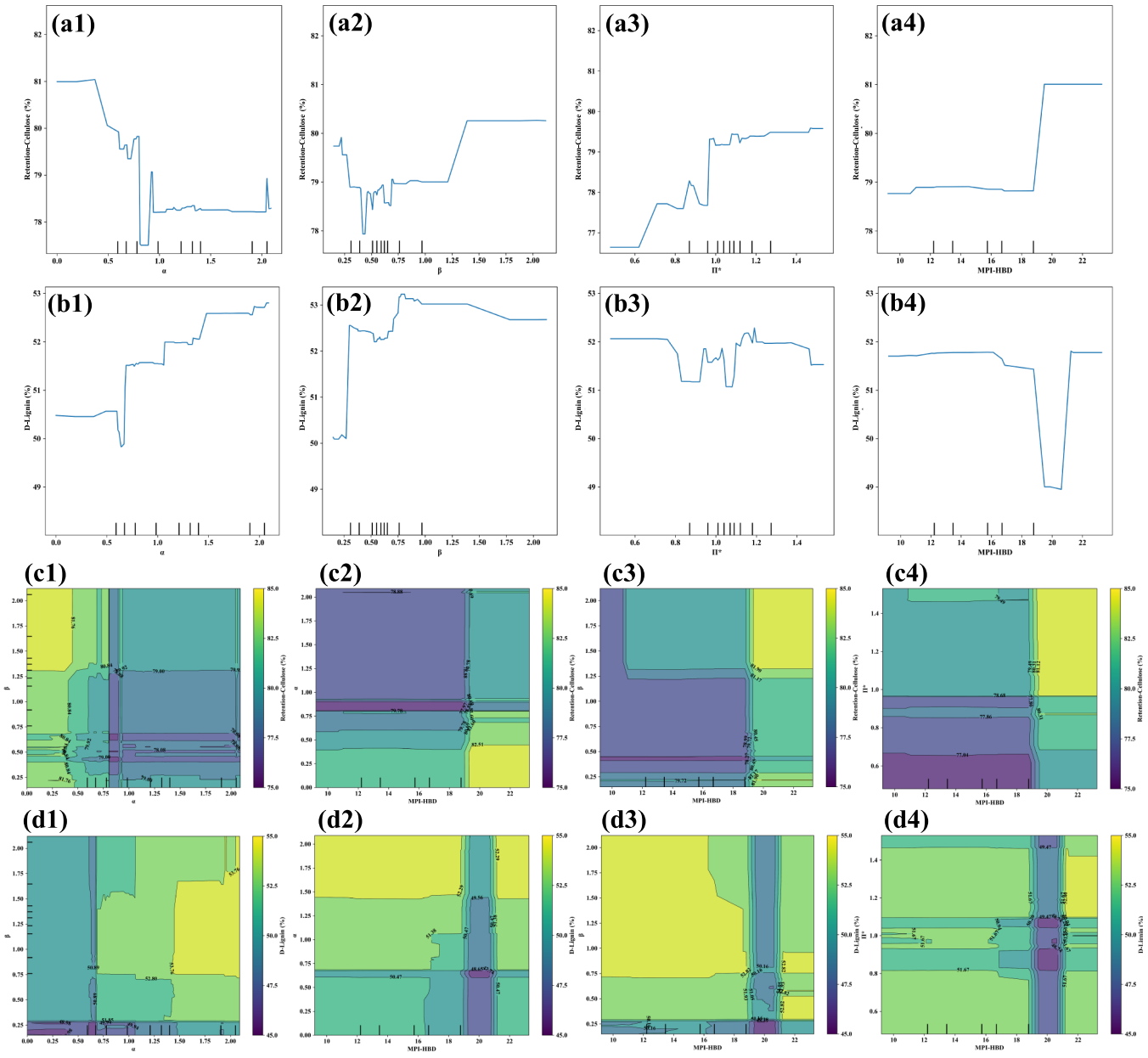


Figure 6. Partial dependence analysis of pretreatment efficiency variables: one-dimensional partial dependence plots of α, β, Π* and MPI-HBD for Retention-Cellulose (a1−a4) and D-Lignin (b1−b4); two-dimensional partial dependence plots of α with β, α with MPI-HBD, β with MPI-HBD and Π* with MPI-HBD for Retention-Cellulose (c1−c4) and D-Lignin (d1−d4).

lignocellulosic biomass is enlightening. However, the ML model itself has a black box problem, and it is difficult to explain the specific mechanism. Therefore, it is necessary to adopt some interpretation methods to increase the interpretability of machine learning. Feature importance analysis of ML models based on SHAP and F-Score theory allows researchers to understand the importance of each variable.[55] The feature importance analysis of the data set was carried out based on the XGBoost model, and the results were displayed in Figure 5.

The *F*-Score for each variable was superimposed by a variable group. It can be seen that for the two dependent variables, the two most important variable groups are the K-T parameter and the polarity parameter of the solvent. For the variable of retention-cellulose, it accounted for 46.1 and 20.8%, respectively. For the D-Lignin variable, it accounted for 41.5 and 22.7%, respectively. This highlights the significance of these two categories of variables. In the SHAP plot, the red dots depict high values, while the blue dots depict low values within the data set. In terms of cellulose retention, the SHAP model indicated that the particle size of biomass and the addition amount of HBD, β, α, and MPI of HBD were highly correlated with it. It is worth noting that although some of the low values of β and α in the figure fall in the positive correlation interval, some of the high values fall in the negative correlation interval. However, most of the β data points have positive SHAP values. This indicates that while β is positively correlated with cellulose retention, α is negatively correlated with cellulose retention. However, this correlation is not linear, it has volatility, and there are extreme values in some ranges.[56] The retention rate of cellulose was also affected by the MPI of the HBD. The model indicated that the higher MPI of the HBD was beneficial to the retention of cellulose.

https://doi.org/10.1021/acssuschemeng.4c01563
ACS Sustainable Chem. Eng. 2024, 12, 7578−7590

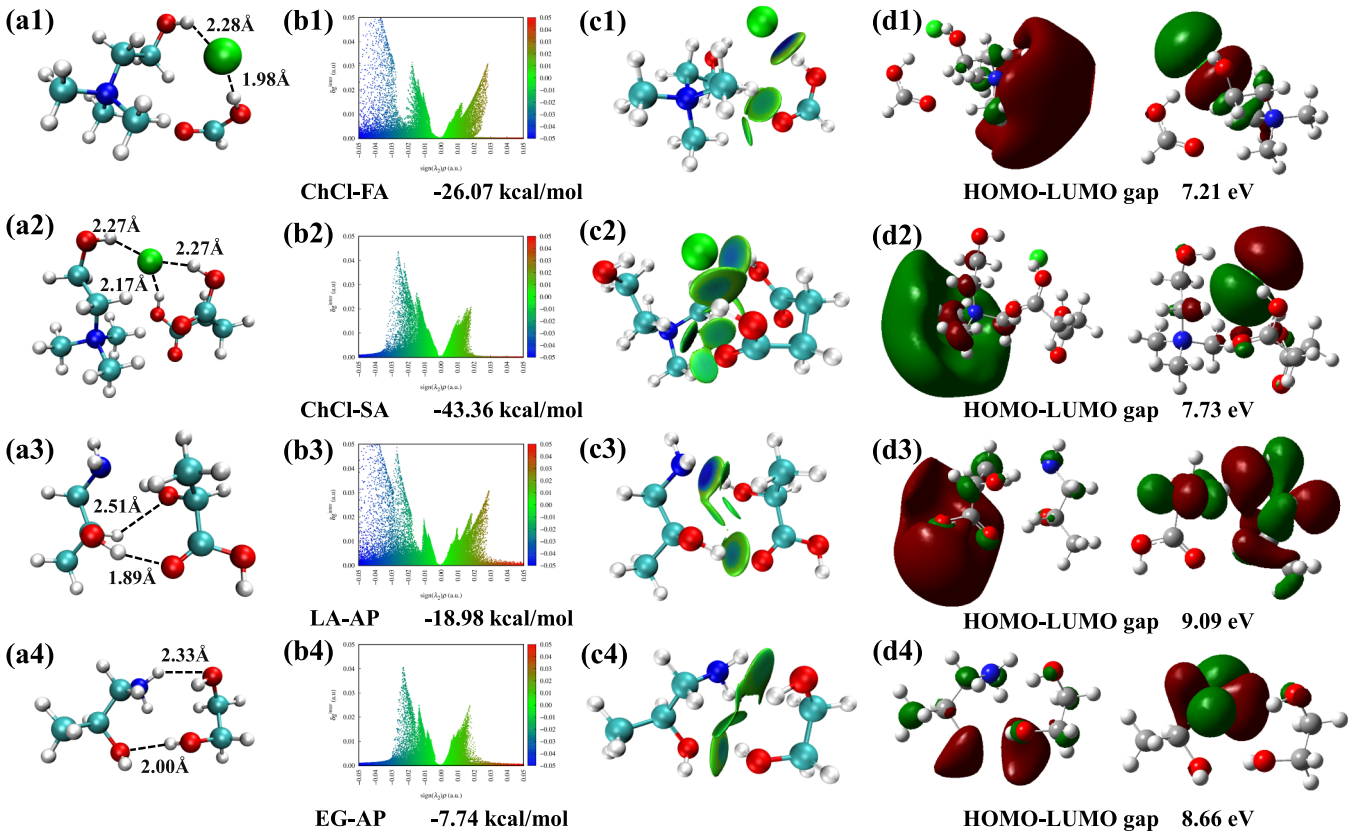


Figure 7. (a1−a4) Optimized structures of ChCl-FA, ChCl-SA, LA-AP and EG-AP; (b1−b4) scatterplot of noncovalent interactions between HBA and HBD (blue: hydrogen bonding; green: van der Waals interaction; red: site-resistance repulsion); (c1−c4) IGMH visual analysis of noncovalent interactions between HBA and HBD (blue: hydrogen bonding; green: van der Waals interaction; red: site-resistance repulsion); (d1−d4) frontier orbital, energy, and HOMO−LUMO gap of DES.

For delignification, the SHAP model indicates that it is more seriously affected by pretreatment conditions, such as temperature and time. At the same time, the degree of correlation with K-T parameters is high. There was a positive correlation with α, a weak correlation with β, and a positive correlation with Π^*. This also proves that DES has a higher α and polarity is more conducive to the removal of lignin.[57] The above conclusions indicate that a screening mechanism for DES can be formed by premeasuring the specific values of the K-T parameters of DES and combining them with the calculated polarity parameters of solvents. The DES screened by this mechanism can retain more cellulose and remove more lignin as much as possible. This mechanism can reduce the time and economic costs of theoretical exploration.

3.5. Exploration of DES Screening Range by Partial Dependence Analysis. A variety of ML model interpretation methods are used to understand the correlation of variables. Partial dependence analysis can quantify the correlation between variables compared with feature importance analysis. Whether the correlation of variables is linear, monotonous, or more complex can be revealed by partial dependence analysis. Based on the XGBoost model, a one-dimensional partial dependence analysis was performed for variables with higher SHAP values in feature importance analysis. The results indicated significant effects of changes in MPI-HBD, α, β, and Π^* on the retention-cellulose and D-lignin. The polar surface area of HBD and the polarity parameters of HBA exhibited minimal influence on the pretreatment efficiency. To analyze the synergistic effect between K-T parameters and polarity parameters, MPI-HBD was combined with α, β, and Π^*, respectively, to perform a two-dimensional partial dependence analysis on pretreatment efficiency. The results are shown in Figure 6. Partial dependence analysis of other minor variables is included in the Supporting Information. The model results show that more than 80% of cellulose can be retained, and more than 50% of lignin can be removed in the process of DES pretreatment by adjusting the values of related variables.

For the cellulose retention rate, the partial dependence analysis results are shown in Figure 6a,c. From the general trend analysis, it can be concluded that the cellulose retention rate decreases with the increase of the α value, while the increase of the β value is beneficial to cellulose retention. DES with higher polarity, especially HBD with MPI values of >20, will allow more cellulose to be retained. According to the results of the two-dimensional partial dependence analysis, the higher values of cellulose retention occurred in the ranges of α less than 0.5 and β more than 1.25. HBD with MPI above 19 and DES with Π^* above 1.0 were able to retain more cellulose components.

For delignification, the partial dependence analysis results are shown in Figure 6b,d. The degree of delignification increased with the increase of α value. β has little effect on delignification when it is above 0.30, but will decrease delignification when it is above 1.30. The effect of solvent polarity on delignification fluctuates greatly, but DES composed of HBD with MPI in the 19 to 21 range was considered to be unfavorable for more lignin to be removed. The two-dimensional partial dependence analysis for the

https://doi.org/10.1021/acssuschemeng.4c01563
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dependent variable D-Lignin is interesting. Corresponding to the results of the one-dimensional partial dependence analysis, α values above 1.5 were considered to be conducive to lignin removal. However, the value of β exists in multiple extreme ranges and it is not necessarily more beneficial for the removal of lignin to have a higher value. DES with high polarity can have a better lignin removal rate even if the value of β is lower than 0.60. The model suggests that the delignification of DES is the highest when the value of α is above 1.50 and the value of β is between 0.75 and 1.75.

Through the partial dependence analysis of the data set, it can be concluded that the relationship between K-T parameters, polarity parameters, and preprocessing efficiency is not simple and linear. The conclusions described above also correspond to the results of the feature importance analysis. The results of the model analysis show that there is a game theory between cellulose retention and lignin removal, and the α and β and polarity parameters should determine a reasonable range. The screening range given by this model can efficiently screen suitable and high-efficiency DES for lignocellulosic biomass pretreatment processes.

3.6. Verification of DES Screening Mechanism by DFT Calculations. The HBA and HBD with higher polarity were selected to form DES. Some other DES are composed of HBA and HBD with lower polarity. The weak interaction between HBA and HBD was investigated by using the density functional theory (DFT) calculation to explore the reaction site. The DES with higher polarity are ChCl-FA and ChCl-SA, while the DES with lower polarity are LA-AP and EG-AP. The DFT calculation results of the above four DES are displayed in Figure 7.

The IGMH analysis method can visualize the weak interactions between molecules. It can be seen from Figure 7c that a large number of hydrogen bonds between HBD and Cl^- on HBA had been formed in the ChCl-based DES system. The distance between the two functional groups that generate hydrogen bonds is about 1.98−2.28 Å, which meets the identification criteria of hydrogen bonds.[37] The hydrogen bond of DES with lower polarity occurs between the hydroxyl and hydroxyl groups or between the hydroxyl and amino groups. The interaction energy of DES with higher polarity is greater, which proves that the noncovalent bonds formed by DES are more stable. Moreover, the vdW force between HBA and HBD is larger, which is generally considered to be the mutual attraction caused by electrostatic force caused by higher molecular polarity.[58] HUMO-LUMO gap values of 4 DES were calculated, and their values were closely related to the overall polarization of DES.[59] The higher the value, the lower the polarizability. The smaller the value, the greater the polarizability. From Figure 7d, it can be seen that DES with the highest polarity has the smallest HUMO-LUMO gap. The above DFT results explain the mechanism that DES with higher polarity has a better pretreatment effect on lignocellulosic biomass from the microscopic point of view. The feasibility of judging the pretreatment effect by the values of the K-T parameter and the polarity parameter of DES is also verified.

4. CONCLUSIONS

A variety of explicable ML models were employed to clarify the intrinsic relationship between K-T parameters and polarity parameters of solvent properties and cellulose retention and delignification during the DES pretreatment of lignocellulose. A method for screening efficient DES was established based on α, β, Π^*, and MPI-HBD. The model predicted that the specific DES could retain more than 80% cellulose and remove more than 50% lignin. The relationship between α and lignin removal rate or β and cellulose retention rate was nonlinear, and the internal relationship of extreme values in some ranges, which were illustrated by the model. The database constructed by using DFT as a descriptor can be further used to predict DES properties. The XGBoost model has the highest accuracy and validates that these descriptors are valid ($R^2 > 0.91$). The method of determining the composition and ratio of DES by the combination of α, β, Π^*, and MPI-HBD parameters is a novel and efficient method. The efficient DES selected by the model was elucidated from the microscopic scale through molecular simulation, and the simulation results showed that a large amount of hydrogen bonds were generated in the system, and these hydrogen bond networks caused better pretreatment effects.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acssuschemeng.4c01563.

Molecular structure and abbreviations; Kamlet−Taft parameter of deep eutectic solvents; polarity parameter of the molecule; predictive performance, accuracy, and hyperparameters of machine learning models; list of references involved in the database; and detailed information on species and composition of lignocellulosic biomass (PDF)

AUTHOR INFORMATION

Corresponding Author

Huanfei Xu − *College of Chemical Engineering, Qingdao University of Science and Technology, Qingdao 266042, PR China; CAS Key Laboratory of Biofuels, Qingdao Institute of Bioenergy and Bioprocess Technology, Chinese Academy of Sciences, Qingdao 266101, PR China;* orcid.org/0000-0002-1188-7220; Email: xuhuanfeixiaotu@163.com

Authors

Hanwen Ge − *College of Chemical Engineering, Qingdao University of Science and Technology, Qingdao 266042, PR China*

Yuekun Bai − *College of Chemical Engineering, Qingdao University of Science and Technology, Qingdao 266042, PR China*

Rui Zhou − *College of Chemical Engineering, Qingdao University of Science and Technology, Qingdao 266042, PR China*

Yaoze Liu − *School of Information Science and Technology, Qingdao University of Science and Technology, Qingdao 266042, PR China*

Jiahui Wei − *College of Chemical Engineering, Qingdao University of Science and Technology, Qingdao 266042, PR China*

Shenglin Wang − *College of Chemical Engineering, Qingdao University of Science and Technology, Qingdao 266042, PR China*

Bin Li − *CAS Key Laboratory of Biofuels, Qingdao Institute of Bioenergy and Bioprocess Technology, Chinese Academy of Sciences, Qingdao 266101, PR China; Shandong Energy*

Institute, Qingdao 266101, PR China; Qingdao New Energy Shandong Laboratory, Qingdao 266101, PR China; orcid.org/0000-0002-8903-3874

Complete contact information is available at: https://pubs.acs.org/10.1021/acssuschemeng.4c01563

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was financially supported by the National Natural Science Foundation of China (22008133).

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https://doi.org/10.1021/acssuschemeng.4c01563
ACS Sustainable Chem. Eng. 2024, 12, 7578−7590